Power Outages: Causes, Impacts, and Data Interpretation

A practical reference for outage records and predictive analytics

Document purpose: This document is prepared as a searchable knowledge-base source for a SmartGrid AI RAG prototype. It is an educational synthesis, not an official ERCOT, utility, or government publication.

What is a power outage?

A power outage is an interruption in electric service to one or more customers. Outages may be planned for maintenance or unplanned because of equipment failure, weather, vegetation, animals, accidents, operational events, or other disturbances. An outage affecting a small distribution circuit is different from a widespread bulk-power system emergency.

An outage record is an observation of an event, not a complete explanation of the event. A dataset may include county, date, duration, number of customers affected, and weather conditions, but it may not contain the root cause, exact asset, restoration sequence, or customer-level detail.

Common causes

Weather-related causes include lightning, high winds, ice, extreme temperatures, flooding, and storms. Infrastructure-related causes include equipment failure, aging assets, protection-system operation, and maintenance activity. External causes can include vehicle collisions, construction damage, vegetation contact, and animal interference.

Multiple causes can occur together. For example, a storm may bring high winds and lightning, damage distribution equipment, and prevent crews from reaching the affected area quickly. A dataset that contains only weather observations and outage duration may show association but cannot prove a specific cause.

Understanding outage metrics

Duration measures how long an outage lasts. Customers affected measures the scale of the interruption. These fields can be highly skewed: many events may be short and affect relatively few customers, while a small number of events may be very large or long-lasting.

Mean customers affected is not necessarily the total number of unique customers interrupted. The same customer may experience more than one interruption, and an aggregated record may represent an average across events or time. Always check the dataset documentation before interpreting a field.

Useful summaries include event counts, median duration, percentile duration, customers affected by county and date, and comparisons across weather conditions. Averages alone can hide rare but severe events.

Data quality checks

Check missing values, duplicate records, inconsistent county names, invalid FIPS codes, impossible coordinates, negative durations, and inconsistent date formats. Confirm whether a row represents an outage event, a county-day summary, or a zone-day summary.

Check the time coverage and geographic coverage. A dataset described as Texas-wide may still omit some utilities, counties, or event types. Avoid describing it as complete unless its documentation supports that claim.

Do not treat synthetic records as historical observations. Synthetic data can help test data pipelines, dashboards, and software integration, but it cannot establish real outage frequencies or prove predictive performance.

How this supports SmartGrid AI

Outage data can be used for exploratory analysis and, when labels and time alignment are appropriate, supervised learning. However, a project predicting grid stress should clearly distinguish a grid-stress label from an actual customer outage label.

Use outage records as historical context and potential explanatory evidence. If the ML model predicts high grid stress, RAG can retrieve relevant guidance about weather, demand response, system reliability, and preparedness. It should not claim that a specific outage will happen unless the model and data are designed to predict that outcome.

Key takeaways

- Outages have many possible causes, and correlation does not prove causation.
- Understand exactly what each row and each metric represents.
- Duration and affected-customer counts are often skewed; inspect medians and percentiles.
- Synthetic data is for software testing, not evidence of real-world outage rates.

References and further reading

• U.S. Department of Energy Event-correlated Outage Dataset in America — https://catalog.data.gov/dataset/event-correlated-outage-dataset-in-america

• NERC Reports — https://www.nerc.com/our-work/reports

Prepared 9 November 2026. Verify current operational information with the linked official sources before making real-world decisions.